EOS Consistency and Catalogue-Relative Separation

DRAFT for Review

Finite-sample guarantees for thresholded behavioral signatures
under active-set, stochastic-coupling, and noise-aware assumptions

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Abstract

This note gives a finite-sample statistical guarantee for EOS as a finite, catalogue-relative behavioral test for stochastic structured-decision pipelines. The result is not a theorem of semantic correctness. It is a theorem of uniform recovery of thresholded behavioral signatures, plus separation of specified adversarial failure modes when the pre-specified catalogue contains margin-separated discriminative probes. The statement makes explicit the pre-specification requirement, active-set conditioning, stochastic coupling used for repeated pipeline runs, finite-catalogue assumptions, and calibration error required for noise-aware EOS.

Context: Empirical Oracle Signature (EOS) in Kelvin

Kelvin evaluates AI pipelines that read structured context and produce a structured decision. It runs controlled perturbations against the actual pipeline: perturbations that should not change the decision test stability, and perturbations that change governing evidence test sensitivity. Kelvin does this without external labels and without an AI judge model.

EOS stands for *Empirical Oracle Signature*. The *oracle* is not an external judge and not a ground-truth label. It is the pre-specified structural expectation encoded by a catalogue item: after applying a controlled transformation T_c, the two pipeline outputs should satisfy the specified relation ρ_c. It is *empirical* because the relation-hold probabilities $p_c(f)$ are estimated from sampled active inputs and repeated pipeline executions. It is a *signature* because EOS records the thresholded set of catalogue items that pass, namely those with $p_c(f) \geq \theta = 1 - \varepsilon$.

Thus EOS is Kelvin's finite, catalogue-relative behavioral fingerprint of a pipeline under controlled perturbation tests. The guarantee is diagnostic and catalogue-relative. It supports evidence-tracking claims inside the tested perturbation catalogue; it is not a proof of full semantic correctness.

1 Setup

Let $(\mathcal{X}, \mathcal{A})$ be a measurable space of typed structured inputs, let $(\mathcal{Y}, \mathcal{B})$ be a measurable output space, and let D be a distribution over $\mathcal{X}$. Let

$$\mathcal{F} = \{f_0, f_1, \ldots, f_A\}$$

be a finite family of stochastic structured-decision pipelines. For each j, let $(\Omega_j, \mathcal{G}_j)$ be the measurable space carrying the internal randomness of pipeline f_j, including model sampling randomness, randomized preprocessing, randomized retrieval, or other randomized pipeline behavior. Each pipeline is represented as a measurable map

$$f_j : (\mathcal{X}, \mathcal{A}) \otimes (\Omega_j, \mathcal{G}_j) \longrightarrow (\mathcal{Y}, \mathcal{B}).$$

EOS evaluates a finite catalogue

$$\mathcal{K} = \{c_1, \ldots, c_M\}, \qquad M = |\mathcal{K}|.$$

Each catalogue item $c \in \mathcal{K}$ consists of:

(i) a measurable transformation $T_c : \mathcal{X} \to \mathcal{X}$;

(ii) a measurable active set $A_c \subseteq \mathcal{X}$ with $D(A_c) > 0$ for every evaluated c;

(iii) a measurable test relation

$$\rho_c : \mathcal{X} \times \mathcal{Y} \times \mathcal{Y} \longrightarrow \{0, 1\}.$$

The usual output-relation formulation $R_c : \mathcal{Y} \times \mathcal{Y} \to \{0, 1\}$ is the special case

$$\rho_c(x, y, y') = R_c(y, y').$$

The slightly more general notation is intentional: noise-aware EOS relations may depend on the input x, for example through an input-dependent replay-noise threshold.

The active set A_c must be defined from input and rule semantics, not from observed pipeline outcomes. If a test has no restricted active set, take $A_c = \mathcal{X}$. If validity is represented by a measurable valid-input set $V \subseteq \mathcal{X}$, assume that T_c maps active valid inputs to valid transformed inputs D-almost surely:

$$D\big(\{x \in A_c \cap V : T_c x \notin V\}\big) = 0.$$

When all points in $\mathcal{X}$ are valid, this condition reduces to measurability of $T_c : \mathcal{X} \to \mathcal{X}$.

For each pair (j, c), fix in advance a row-level probability kernel

$$\Gamma_{j,c} : (\mathcal{X}, \mathcal{A}) \rightsquigarrow (\Omega_j \times \Omega_j, \mathcal{G}_j \otimes \mathcal{G}_j).$$

Equivalently, $\Gamma_{j,c}(x)$ is a probability measure on $(\Omega_j \times \Omega_j, \mathcal{G}_j \otimes \mathcal{G}_j)$, measurably indexed by x. This kernel specifies how the two stochastic executions in the comparison are coupled. The default case is independent internal randomness for $f_j(x)$ and $f_j(T_c x)$, but common-random-number or deterministic-replay couplings are also allowed if they are fixed before evaluation. The theorem applies to the coupling actually used; changing the coupling changes the population quantity being estimated.

For $X \sim D(\cdot \mid A_c)$ and $(W_1, W_2) \sim \Gamma_{j,c}(X)$, define the active-set relation-hold probability

$$p_c(f_j) = \mathrm{P}\big(\rho_c(X, f_j(X, W_1), f_j(T_c X, W_2)) = 1 \mid X \in A_c\big).$$

The conditioning notation is shorthand for the sampling procedure just described: first draw X from the active distribution $D(\cdot \mid A_c)$, then draw the coupled internal randomness according to $\Gamma_{j,c}(X)$.

Fix an EOS tolerance $\varepsilon \in (0, 1)$ and write

$$\theta = 1 - \varepsilon.$$

The true EOS signature of pipeline f_j is

$$\Sigma_\varepsilon(f_j) = \{c \in \mathcal{K} : p_c(f_j) \geq \theta\}.$$

For each evaluated pair (j, c), let

$$X_1^{(j,c)}, \ldots, X_{n_{j,c}}^{(j,c)} \overset{\text{iid}}{\sim} D(\cdot \mid A_c).$$

For each row, draw

$$(W_{1,i}^{(j,c)}, W_{2,i}^{(j,c)}) \sim \Gamma_{j,c}(X_i^{(j,c)})$$

independently across rows for every fixed (j, c). Define

$$Z_{i,j,c} = \rho_c(X_i^{(j,c)}, f_j(X_i^{(j,c)}, W_{1,i}^{(j,c)}), f_j(T_c X_i^{(j,c)}, W_{2,i}^{(j,c)})),$$

so that

$$\widehat{p}_c(f_j) = \frac{1}{n_{j,c}} \sum_{i=1}^{n_{j,c}} Z_{i,j,c}.$$

The empirical EOS signature is

$$\widehat{\Sigma}_\varepsilon(f_j) = \{c \in \mathcal{K} : \widehat{p}_c(f_j) \geq \theta\}.$$

Dependence across different catalogue items or different pipelines is not needed for the results below. The proofs use only per-(j, c) row independence and union bounds.

Assumption 1 (Pre-specification)**.** The catalogue $\mathcal{K}$, transformations T_c, active sets A_c, relations ρ_c, tolerance ε, threshold θ, stochastic coupling kernels $\Gamma_{j,c}$, family of pipelines to be compared, specified adversaries, calibration protocol, and any claimed margin parameter λ are fixed independently of the evaluation samples and evaluation outcomes. If any of these are chosen adaptively after inspecting evaluation outcomes, the claims below require an additional valid adaptive-data-analysis or holdout correction.

Remark 1 (Active counts)**.** The theorems below are stated for active examples sampled directly from $D(\cdot \mid A_c)$ with fixed active sample sizes $n_{j,c}$. If active examples are instead obtained by filtering raw IID draws from D, the same guarantees hold conditional on the realized active counts satisfying the stated lower bounds. Proposition 1 gives one simple unconditional conversion from raw sample size to active sample size.

2 Single-pipeline consistency

Theorem 1 (Active-set EOS consistency for one stochastic pipeline)**.** *Fix a pipeline f. Assume:*

(i) $\mathcal{K}$ is finite with $|\mathcal{K}| = M$;

(ii) the pre-specification assumption holds;

(iii) for each $c \in \mathcal{K}$, the evaluated examples are IID from $D(\cdot \mid A_c)$ and the row-level internal randomness is independent across rows;

(iv) every T_c maps active valid inputs to valid transformed inputs D-almost surely;

(v) there is a boundary margin $\lambda > 0$ such that, for every $c \in \mathcal{K}$,

$$|p_c(f) - \theta| \geq \lambda.$$

Let

$$n_{\min} = \min_{c \in \mathcal{K}} n_c.$$

If

$$n_{\min} \geq \frac{1}{2\lambda^2} \log \frac{2M}{\delta},$$

then

$$\mathrm{P}\big(\widehat{\Sigma}_\varepsilon(f) = \Sigma_\varepsilon(f)\big) \geq 1 - \delta.$$

If active counts are random because they arise from filtering raw IID samples, the same statement holds conditional on the realized $n_{\min}$ satisfying the displayed bound.

Proof. For fixed $c \in \mathcal{K}$, the variables $Z_{1,c}, \ldots, Z_{n_c,c}$ are IID Bernoulli random variables with mean $p_c(f)$, and

$$\widehat{p}_c(f) = \frac{1}{n_c} \sum_{i=1}^{n_c} Z_{i,c}.$$

By Hoeffding's inequality,

$$\mathrm{P}\big(|\widehat{p}_c(f) - p_c(f)| \geq \lambda\big) \leq 2e^{-2n_c\lambda^2} \leq 2e^{-2n_{\min}\lambda^2}.$$

A union bound over the M catalogue items gives

$$\mathrm{P}\big(\exists c \in \mathcal{K} : |\widehat{p}_c(f) - p_c(f)| \geq \lambda\big) \leq 2Me^{-2n_{\min}\lambda^2}.$$

On the complementary event, every empirical probability lies on the same side of θ as its true probability. If $p_c(f) \geq \theta + \lambda$, then $\widehat{p}_c(f) > \theta$; if $p_c(f) \leq \theta - \lambda$, then $\widehat{p}_c(f) < \theta$. Hence $\widehat{\Sigma}_\varepsilon(f) = \Sigma_\varepsilon(f)$. Setting $2Me^{-2n_{\min}\lambda^2} \leq \delta$ proves the claim. □

3 Uniform consistency over a finite family

Theorem 2 (Uniform active-set EOS consistency over finite pipelines)**.** *Let $\mathcal{F} = \{f_0, f_1, \ldots, f_A\}$ be finite. Assume the conditions of Theorem 1 hold for every $f_j \in \mathcal{F}$, and assume the uniform margin condition*

$$|p_c(f_j) - \theta| \geq \lambda \qquad \text{for all } j = 0, \ldots, A,\ c \in \mathcal{K}.$$

Let

$$n_{\min} = \min_{0 \leq j \leq A,\ c \in \mathcal{K}} n_{j,c}.$$

If

$$n_{\min} \geq \frac{1}{2\lambda^2} \log \frac{2M(A+1)}{\delta},$$

then

$$\mathrm{P}\Big(\forall j \in \{0, \ldots, A\},\ \widehat{\Sigma}_\varepsilon(f_j) = \Sigma_\varepsilon(f_j)\Big) \geq 1 - \delta.$$

If active counts are random because they arise from filtering raw IID samples, the same statement holds conditional on the realized $n_{\min}$ satisfying the displayed bound.

Proof. Apply the proof of Theorem 1 to every pipeline-catalogue pair (j, c). There are $M(A+1)$ such estimates. Hoeffding's inequality and a union bound give failure probability at most

$$2M(A+1)e^{-2n_{\min}\lambda^2}.$$

Setting this quantity at most δ proves the claim. □

4 Converting raw samples to active samples

Proposition 1 (One sufficient raw-sample bound)**.** *Let $n_\star$ be an active-sample lower bound required by one of the preceding theorems. Suppose active examples are produced by filtering raw IID draws from D, and suppose every active probability being controlled is at least*

$$\alpha_{\min} = \min_c D(A_c) > 0.$$

Let L be the number of active counts that must simultaneously meet the lower bound. For example, $L = M$ for a single pipeline if one raw pool is used per catalogue item, and $L \leq M(A+1)$ for the fully separate uniform-family case. If

$$N \geq \max\left\{\frac{2n_\star}{\alpha_{\min}}, \frac{8}{\alpha_{\min}} \log \frac{L}{\delta_{\mathrm{act}}}\right\},$$

then, with probability at least $1 - \delta_{\mathrm{act}}$, all L realized active counts are at least $n_\star$. Combining this event with Theorem 1 or Theorem 2 gives total failure probability at most $\delta + \delta_{\mathrm{act}}$.

Proof. For each controlled active count, $n \sim \mathrm{Binomial}(N, \alpha)$ with $\alpha \geq \alpha_{\min}$ and mean $\mu = N\alpha \geq N\alpha_{\min}$. The displayed condition implies $\mu \geq 2n_\star$, so $n_\star \leq \mu/2$. Chernoff's bound gives

$$\mathrm{P}(n < n_\star) \leq \mathrm{P}(n < \mu/2) \leq e^{-\mu/8} \leq e^{-N\alpha_{\min}/8} \leq \frac{\delta_{\mathrm{act}}}{L}.$$

A union bound over the L counts proves the result. □

5 Separation of rule-tracking from specified adversaries

Let $f_0 = f_{\text{track}}$ and let $f_1, \ldots, f_A$ be specified adversaries.

Theorem 3 (Catalogue-relative separation)**.** *Suppose that for every adversary f_j, $j \geq 1$, there exists a discriminative catalogue item $c_j \in \mathcal{K}$ such that either*

$$p_{c_j}(f_0) \geq \theta + \lambda \quad \textit{and} \quad p_{c_j}(f_j) \leq \theta - \lambda,$$

or the reverse inequalities hold:

$$p_{c_j}(f_0) \leq \theta - \lambda \quad \textit{and} \quad p_{c_j}(f_j) \geq \theta + \lambda.$$

Let $n_{\min}^{\text{sep}}$ be the minimum active sample size among the estimates

$$\widehat{p}_{c_j}(f_0),\ \widehat{p}_{c_j}(f_j), \qquad j = 1, \ldots, A,$$

counting repeated estimates only once if the same estimate is reused. If

$$n_{\min}^{\text{sep}} \geq \frac{1}{2\lambda^2} \log \frac{4A}{\delta},$$

then

$$\mathrm{P}\Big(\widehat{\Sigma}_\varepsilon(f_0) \neq \widehat{\Sigma}_\varepsilon(f_j) \textit{ for every } j = 1, \ldots, A\Big) \geq 1 - \delta.$$

A sufficient but generally stronger condition is the uniform recovery bound from Theorem 2:

$$n_{\min} \geq \frac{1}{2\lambda^2} \log \frac{2M(A+1)}{\delta}.$$

Proof. For each adversary f_j, it is enough to estimate the two probabilities $p_{c_j}(f_0)$ and $p_{c_j}(f_j)$ within error strictly less than λ. Hoeffding's inequality and a union bound over at most $2A$ required estimates give failure probability at most

$$4Ae^{-2n_{\min}^{\text{sep}}\lambda^2}.$$

On the complementary event, every discriminative item c_j remains on opposite sides of θ for f_0 and f_j. Hence c_j belongs to exactly one of the two empirical signatures, so those empirical signatures differ. Setting the displayed failure probability at most δ proves the result. □

6 Observable failure mass for stochastic adversaries

The separation theorem is conditional on the existence of discriminative catalogue items. For stochastic clause-drop, clause-corruption, or rule-corruption adversaries, this requires enough observable net failure mass under the relevant active distribution.

Fix a catalogue item c. Couple the tracker and adversary on a common probability space with the correct marginals for their individual EOS evaluations, and define their relation indicators Z_0 and Z_a. Then

$$p_c(f_{\text{track}}) - p_c(f_{\text{adv}}) = \mathrm{P}(Z_0 = 1, Z_a = 0 \mid X \in A_c) - \mathrm{P}(Z_0 = 0, Z_a = 1 \mid X \in A_c).$$

Thus a harmful corruption probability alone is not enough; compensating flips in the opposite direction must be absent or bounded.

Lemma 1 (Sufficient observable net failure mass)**.** *Fix $c \in \mathcal{K}$ and define the net relation-loss mass*

$$\Delta_c^{\text{loss}} = p_c(f_{\text{track}}) - p_c(f_{\text{adv}}).$$

If

$$\Delta_c^{\text{loss}} \geq p_c(f_{\text{track}}) - (\theta - \lambda),$$

then

$$p_c(f_{\text{adv}}) \leq \theta - \lambda.$$

If also

$$p_c(f_{\text{track}}) \geq \theta + \lambda,$$

then c is a margin-separated discriminative catalogue item and Theorem 3 applies.

In the clean special case $p_c(f_{\text{track}}) = 1$ and $\theta = 1 - \varepsilon$, the sufficient condition becomes

$$\Delta_c^{\text{loss}} \geq \varepsilon + \lambda.$$

This clean case can certify the tracking side with margin only when $\lambda \leq \varepsilon$, because $1 \geq \theta + \lambda = 1 - \varepsilon + \lambda$ is otherwise impossible.

Proof. The first displayed condition implies

$$p_c(f_{\text{adv}}) = p_c(f_{\text{track}}) - \Delta_c^{\text{loss}} \leq \theta - \lambda.$$

Together with $p_c(f_{\text{track}}) \geq \theta + \lambda$, this places the tracker and adversary on opposite sides of the threshold with margin λ. The clean-case expression follows by substituting $p_c(f_{\text{track}}) = 1$ and $\theta = 1 - \varepsilon$. □

Remark 2 (Interpreting $p_{\text{attack}}\beta_c$)**.** If E_c denotes a corruption event and H_c denotes the event that the corruption is harmful and decision-revealing for catalogue item c, then

$$\mathrm{P}(E_c \cap H_c \mid X \in A_c) = p_{\text{attack}}\beta_c,$$

provided

$$p_{\text{attack}} = \mathrm{P}(E_c \mid X \in A_c) \qquad \text{and} \qquad \beta_c = \mathrm{P}(H_c \mid E_c, X \in A_c).$$

No independence assumption is needed for this identity. However, this product contributes to separation only through net relation loss after subtracting any compensating mass that makes the adversary satisfy the relation in cases where the tracker would not.

7 Noise-aware EOS

Noise-aware EOS replaces a raw relation by a relation that ignores changes not exceeding a pre-specified estimate of the pipeline's own replay noise. In the present notation, an oracle noise-aware relation is a measurable indicator

$$\rho_c^{\Omega} : \mathcal{X} \times \mathcal{Y} \times \mathcal{Y} \longrightarrow \{0, 1\}.$$

For example, one may have

$$\rho_c^{\Omega}(x, y, y') = 1 \quad \Longleftrightarrow \quad m_c(x, y, y') \geq \tau_c(x) + \kappa_c,$$

where $m_c : \mathcal{X} \times \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}$ is a measurable margin statistic, $\tau_c(x)$ is an oracle replay-noise threshold, and κ_c is a pre-specified signal margin. In evaluation, this relation is applied at $y = f_j(X, W_1)$ and $y' = f_j(T_c X, W_2)$. Define the oracle active-set noise-aware probability

$$p_c^{\Omega}(f_j) = \mathrm{P}\Big(\rho_c^{\Omega}(X, f_j(X, W_1), f_j(T_c X, W_2)) = 1 \mid X \in A_c\Big).$$

The oracle noise-aware signature is

$$\Sigma_\varepsilon^\Omega(f_j) = \{c \in \mathcal{K} : p_c^\Omega(f_j) \geq \theta\}.$$

In practice, τ_c or the full relation ρ_c^Ω may be estimated from an independent calibration object C, producing a plug-in relation $\widehat{\rho}_{C,c}^\Omega$. Conditional on the realized calibration object C, define the true plug-in probability

$$\widetilde{p}_{C,c}^\Omega(f_j) = \mathrm{P}\Big(\widehat{\rho}_{C,c}^\Omega(X, f_j(X, W_1), f_j(T_c X, W_2)) = 1 \mid X \in A_c, C\Big).$$

Let $\widehat{p}_{C,c}^\Omega(f_j)$ be the corresponding empirical average on evaluation samples independent of C, and define the empirical plug-in signature

$$\widehat{\Sigma}_{C,\varepsilon}^\Omega(f_j) = \{c \in \mathcal{K} : \widehat{p}_{C,c}^\Omega(f_j) \geq \theta\}.$$

Assumption 2 (Uniform plug-in relation error)**.** There exist $\eta \geq 0$ and $\delta_{\mathrm{cal}} \in [0, 1)$ such that, with probability at least $1 - \delta_{\mathrm{cal}}$ over the calibration randomness,

$$\left|\widetilde{p}_{C,c}^\Omega(f_j) - p_c^\Omega(f_j)\right| \leq \eta \qquad \text{for every } j = 0, \ldots, A,\ c \in \mathcal{K}.$$

The calibration object C is independent of the evaluation samples used to estimate plug-in probabilities. Conditional on C, the evaluation indicators are independent across rows for each fixed (j, c).

Remark 3 (A sufficient anti-boundary condition)**.** A common sufficient condition for Assumption 2 is the following uniform condition: with probability at least $1 - \delta_{\mathrm{cal}}$ over the calibration randomness, for every $j = 0, \ldots, A$ and every $c \in \mathcal{K}$, under the coupled evaluation law $X \sim D(\cdot \mid A_c)$ and $(W_1, W_2) \sim \Gamma_{j,c}(X)$, the estimated threshold $\widehat{\tau}_{C,c}(x)$ is uniformly within ν of $\tau_c(x)$ on A_c, and the oracle margin statistic has small anti-boundary mass:

$$\mathrm{P}\big(\big|m_c(X, f_j(X, W_1), f_j(T_c X, W_2)) - [\tau_c(X) + \kappa_c]\big| \leq \nu \mid X \in A_c\big) \leq \eta.$$

This condition is not automatic. It must hold uniformly over the claimed family and catalogue, or else it must be verified, bounded, or stated as an explicit modelling assumption with the appropriate non-uniform constants.

Theorem 4 (Noise-aware active-set EOS consistency)**.** *Assume the uniform plug-in relation error assumption holds with parameters $(\eta, \delta_{\mathrm{cal}})$. Assume also that the oracle noise-aware probabilities have margin λ:*

$$|p_c^\Omega(f_j) - \theta| \geq \lambda \qquad \textit{for all } j = 0, \ldots, A,\ c \in \mathcal{K},$$

with $\lambda > \eta$. Let

$$n_{\min} = \min_{0 \leq j \leq A,\ c \in \mathcal{K}} n_{j,c}$$

be the minimum active evaluation sample size for the plug-in estimates. If

$$n_{\min} \geq \frac{1}{2(\lambda - \eta)^2} \log \frac{2M(A+1)}{\delta_{\mathrm{eval}}},$$

then

$$\mathrm{P}\Big(\forall j = 0, \ldots, A, \widehat{\Sigma}_{C,\varepsilon}^\Omega(f_j) = \Sigma_\varepsilon^\Omega(f_j)\Big) \geq 1 - \delta_{\mathrm{eval}} - \delta_{\mathrm{cal}}.$$

If active counts are random because they arise from filtering raw IID samples, the same statement holds conditional on the realized $n_{\min}$ satisfying the displayed bound.

Proof. On the calibration-success event from Assumption 2, every plug-in probability differs from its oracle probability by at most η. Therefore every plug-in probability remains on the same side of θ as its oracle probability, with effective margin at least $\lambda - \eta$.

Conditional on the calibration object C and on calibration success, the empirical plug-in averages are averages of bounded independent indicators for each fixed (j, c). Hoeffding's inequality and a union bound over $M(A+1)$ pairs give conditional evaluation failure probability at most

$$2M(A+1)e^{-2n_{\min}(\lambda-\eta)^2}.$$

The displayed sample-size condition makes this at most δ_{eval}. Adding the calibration failure probability δ_{cal} proves the result. □

8 Catalogue-relative identifiability

The guarantee is catalogue-relative. It does not imply semantic correctness outside the tested catalogue algebra.

Let Π be a target partition of $\mathcal{F}$ into behavioral classes to be distinguished, for example

$$\{\text{rule-tracking},\ \text{rule-blind},\ \text{constant},\ \text{wrong-static},\ \text{stochastically broken}\}.$$

Definition 1 (Separating catalogue for a target partition)**.** The catalogue $\mathcal{K}$ is (ε, λ)-separating for Π over $\mathcal{F}$ if, whenever $f, g \in \mathcal{F}$ belong to different cells of Π, there exists $c \in \mathcal{K}$ such that either

$$p_c(f) \geq \theta + \lambda \quad \text{and} \quad p_c(g) \leq \theta - \lambda,$$

or the reverse inequalities hold.

Corollary 1 (No merging of separated behavior classes)**.** *If $\mathcal{K}$ is (ε, λ)-separating for Π over $\mathcal{F}$, then, under the active-sample bound in Theorem 2, empirical EOS signatures do not merge distinct cells of Π with probability at least $1 - \delta$.*

Proof. Theorem 2 recovers all true thresholded signatures. Since $\mathcal{K}$ separates every pair of pipelines in different cells of Π, their true signatures differ. Hence their empirical signatures also differ on the recovery event. □

9 Consolidated statement

For a finite family of stochastic structured-decision pipelines, a finite pre-specified catalogue of valid schema-derived transformations, pre-specified active sets, pre-specified stochastic execution couplings, and measurable relations, EOS consistently estimates each pipeline's catalogue-relative behavioral signature with probability at least $1 - \delta$, provided the true active-set relation probabilities are separated from the EOS acceptance boundary by margin λ and

$$n_{\min} \geq \frac{1}{2\lambda^2} \log \frac{2M(A+1)}{\delta}.$$

If $A_c = \mathcal{X}$ for all catalogue items and the same N examples are used throughout, this reduces to the full-distribution bound with $n_{\min} = N$.

If the catalogue contains at least one margin-separated discriminative transformation-relation item for each specified adversarial failure mode, then EOS separates rule-tracking pipelines from those adversaries with the same concentration guarantees.

For noise-aware EOS, the same result holds with effective margin

$$\lambda_{\text{eff}} = \lambda - \eta,$$

where η bounds the probability-level error introduced by threshold calibration and near-boundary examples. The corresponding active-sample bound is

$$n_{\min} \geq \frac{1}{2(\lambda - \eta)^2} \log \frac{2M(A+1)}{\delta_{\text{eval}}}, \qquad \lambda > \eta,$$

with additional calibration failure probability δ_{cal}.

10 What the theorem proves and does not prove

It proves:

- statistical consistency of EOS signatures for a finite, pre-specified catalogue;
- active-set consistency when applicability subsets are defined from input and rule semantics before evaluation;
- judge-free estimation in the sense that no external labels are needed to compute the signature;
- separation of specified adversarial failure modes when the catalogue contains margin-separated discriminative probes;
- noise-aware validity under explicit calibration-error and anti-boundary-mass assumptions;
- catalogue-relative behavioral identifiability for target classes separated by the catalogue.

It does not prove:

- full semantic correctness of a pipeline;
- detection of failures absent from the transformation-relation catalogue;
- validity for catalogues, adversaries, thresholds, stochastic couplings, or margins selected after inspecting evaluation outcomes, unless a further adaptive-data-analysis correction is supplied;
- separation when true probabilities lie too close to the threshold $1 - \varepsilon$;
- detection of stochastic bugs whose observable net failure mass is below the required margin;
- equivalence outside the catalogue-observable behavior tested by EOS.

11 Numerical scale

For example, with $M = 100$ catalogue items and $\delta = 0.05$, the single-pipeline active-sample bound, rounded up to the next integer, is:

Margin λ	Required active samples per item $n_{\min}$
0.10	415
0.05	1,659
0.02	10,368

These are worst-case Hoeffding bounds. They are conservative but distribution-free. For active-set tests, the relevant quantity is the effective active count for the catalogue item, not merely the total number of raw inputs sampled. For the uniform-family theorem, replace M by $M(A+1)$ in the logarithmic factor.